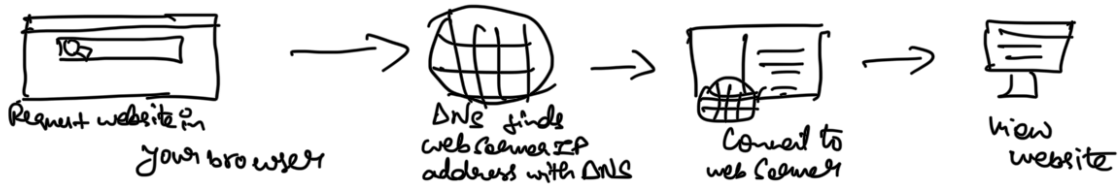
Putting it all together:-

when you request a webpage in your browser. To summarize, when you request a website, your computer need to know server's IP address it needs to talk for this it uses DNS. your computer then talks to web server using a special set of commands called HTTP protocol; the webserver then returns HTML, CSS, Javascript, Images etc. which your browser then uses to correctly format and display the website to you.



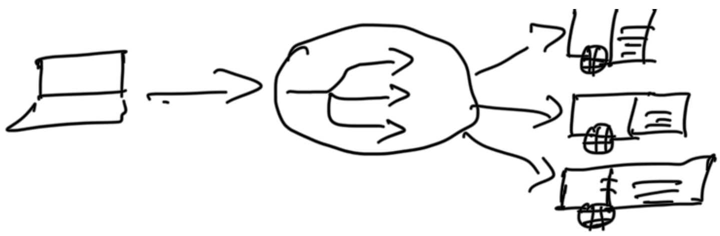
Task 2:- other components

1.) Load Balancers :- when website traffic starts getting quite large or is running an application that needs to have high availability, one web server might no longer do the job.

-> load Balancer provide 2 main features

1. ensuring high traffic website can handle the load 2. Providing a failover if server becomes unresponsive.

* when you request a website with load balancer, the loadbalancer will receive your request first and then forward it to one of multiple servers behind it.
* The load balancer uses different algorithm to help deal with which server is best to deal with request. Ex:- Round Robin, which sends it to each server in turn or weighted, which checks how many requests a server is currently dealing with and sends it to least busy server.
* load balancers also provide periodic checks with each server to ensure they are running currently this is called health check. If server doesn't respond appropriately or doesn't respond the load balancer will stop sending traffic until it responds appropriately again.



CDN (Content Delievery Network):-

CDN is excellent resource for cutting down traffic to a busy website. It allows you to host static files from website, such as JS, CSS, Images, videos & host them across thousand of servers all over the world. When user request one of hosted file, CDN works out where nearest server is physically located and sends request there instead of potentially the other side of world.

Databases:-

Often websites need a way of storing information for their users. Webservers can communicate with databases to store and recall data from them.

* Databases can range from just simple plain text file up to complex clusters of multiple servers providing speed and resilience. Some common databases are MySQL, MSSQL, MongoDB, postgres and more.

WAF (Web Application Firewall):-

* A WAF sits behind your web request and web server; its primary purpose is to protect webserver from hacking or denial of service attacks. It analyses web requests for common attack techniques, whether request is from real browser rather than bot. Also checks if excessive amount of web requests are being sent by utilising something called rate limiting, which allows certain amount of requests from IP per second. If request is deemed a potential attack, it will be dropped and never sent to webserver.

Quiz:-

using.

1) what can be used to host static files and speed up clients visit to website?

A) CDN

2) what does a load balancer perform to make sure a host is still alive?

A) Health Check

3) what can be used to help against hacking a website?

A) WAF